Mariano Marcos State University
THE Impact Rankings 2027 · SDG 15 Life on Land

STORYBOARD · INDICATOR

15.3.5

Collaboration for shared land ecosystems

2025 EVIDENCE

Evidence year **2025**
Data submission due **9 Nov 2026**
6 × 2025 evidence · 2 × continuing

15 LIFE ON LAND

THE ASKS

Collaborate with the local community to maintain shared land ecosystems.

MMSU'S COMMITMENT

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

STORYLINE

Shared land, shared work: LGUs, DENR, a service club, a foundation and barangays planting and protecting together.

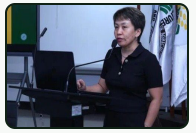
2025 EVIDENCE
Dated 2025 items support this indicator.

THE FILMSTRIP · 8 FRAMES


1


2


3


4


5


6


7


8

AT A GLANCE

1,750
seedlings planted in Pasuquin

4
agreement parties

3
mangrove plantings in 2025



Signing of the memorandum of agreement among MMSU, TMP Foundation, DENR and LGU-Pasuquin, 29 November 2025. (Photo: MMSU STRATCOM)



MMSU, TMP Foundation, DENR and LGU-Pasuquin officials at the agreement signing, 29 November 2025. (Photo: MMSU STRATCOM)

29 NOV 2025 · BRGY. DAVILA, PASUQUIN, ILOCOS NORTE

MMSU, TMP Foundation, DENR and Pasuquin LGU Sign Agreement to Conserve a Coastal Forest

1. **THE AGREEMENT**
   MMSU, the Toyota Motor Philippines Foundation, DENR and LGU-Pasuquin signed a memorandum of agreement to strengthen coastal forest conservation; President Manzano signed for MMSU.

2. **THE PLANTING**
   At the degraded 2-hectare National Greening Program site, about 110 volunteers (incl. MMSU College of Industrial Technology, DENR-CENRO Bangui, the MENRO and residents) planted 1,750 seedlings.

3. **THE COMMITMENT**
   Over two years the partners will plant 3,500 seedlings of native lapis-lapis and arinaya, for shared stewardship, rehabilitated coastal systems and community resilience.

4	110	1,750	3,500
signing parties	volunteers	seedlings planted	two-year target

LENS FOR 15.3.5

A four-party agreement with a degraded coastal forest as the shared ecosystem.

Source: MMSU News, “MMSU ties up with Toyota and DENR for coastal preservation in Pasuquin town,” 4 December 2025 · The Manila Times, “TMP Foundation adopts mangrove forest for climate mitigation,” 10 December 2025



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

6 DEC 2025 · BRGY. PAGSANAHAN, BADOC, ILOCOS NORTE

MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand

1. **THE PLANTING**
   MMSU, the Rotary Club of Laoag Sunshine and LGU-Badoc planted 280 mangrove propagules to restore a stand extensively damaged by past typhoons.

2. **THE GOAL**
   To ensure the continuity of ecological succession at the site.

3. **WHO LED IT**
   President Manzano with Badoc's Hon. Virgilio M. Calajate and Rotary's Atty. Zsa Zsa Raval; MMSU's Vice President for Resource Generation and directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs also took part.

280
mangrove propagules

3
partners

LENS FOR 15.3.5

A community and service-club partnership restoring a shared coastal stand.

Source: MMSU News, "MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town," 11 December 2025



Mangrove planting for World Environment Day, Brgy. Maglaoi, Currimao, June 2025. (Photo: MMSU STRATCOM)



MMSU, LGU Currimao, DENR and fisherfolk volunteers at the planting site, June 2025. (Photo: MMSU STRATCOM)

JUNE 2025 · BRGY. MAGLAOI, CURRIMAO, ILOCOS NORTE

MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day

1. **THE EVENT**
   For World Environment Day, MMSU planted 250 mangrove propagules along the Maglaoi coast to restore degraded shorelines and strengthen climate resilience. The Sustainability and Futures Thinking Office led it with LGU Currimao.

2. **WHO TOOK PART**
   Faculty, education support personnel and students joined DENR representatives, barangay officials, local fisherfolk organisations and community members.

3. **WHY MANGROVES**
   SFT Director Dr. Arlene Gonzales: mangroves filter plastic waste before it reaches the sea, absorb carbon, protect against erosion and shelter marine life.

250
mangrove propagules

LENS FOR 15.3.5

MMSU, LGU Currimao, DENR and fisherfolk organisations planting together.

Source: MMSU News, “MMSU plants 250 mangrove propagules in Currimao,” 11 June 2025



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.



MMSU and DENR Region 1 officials during the dialogue at ILAARRDEC.

2025 · ILAARRDEC, ILOCOS NORTE · DATE TO CONFIRM

MMSU and DENR Region 1 Strengthen Partnership for Mangrove Conservation

1. **THE DIALOGUE**
   MMSU and DENR Region 1 convened a dialogue to advance mangrove conservation and coastal resilience in the Ilocos Region through science-based, community-centred approaches.

2. **THE PLEDGE**
   DENR officials pledged technical support and active participation in restoration, monitoring and evaluation, centred on the DOST-PCAARRD NbS mangrove project led by Dr. Arlene Gonzales.

3. **THE FOLLOW-THROUGH**
   The same project brought representatives of Currimao, Pasuquin, Badoc and Burgos, with DENR-PENRO, to MMSU's mangrove propagation seminar-workshop on 27–28 Oct 2025.

4
coastal LGUs engaged

LENS FOR 15.3.5

A formal partnership (MOU of October 2024) with the national agency responsible for the land.

Source: MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking · MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

OCTOBER 2025 · SAN NICOLAS AND VINTAR, ILOCOS NORTE

Indigenous Food Plant Foodscapes Established with Vintar and San Nicolas Partners

1. **THE PARTNERSHIP**
   The Extension and Research Directorates partnered with LGUs and schools to establish Community Foodscapes and Seedbanks for Indigenous Food Plants at Eladio V. Barangan Memorial Elementary School.
2. **THE PARTNERS**
   Officials of Barangays San Roque and Parut (Vintar), the Municipal Agriculture Office of Vintar, extension workers, school leaders and community representatives, the second batch of partner sites.
3. **SHARED STEWARDSHIP**
   Communities maintain the foodscapes as shared seedbanks and learning gardens with MMSU planting materials and technical assistance; they join existing sites in Sarrat and Batac.

2nd
batch of partner sites

3
sites incl. Sarrat, Batac

LENS FOR 15.3.5

Barangays and schools maintain shared seedbanks and learning gardens.

Source: MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 · MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

2025 · BRGY. AGLIPAY, PINILI, ILOCOS NORTE

A New Bambusetum in Barangay Aglipay, Pinili

1. **THE TRAINING**
   In a two-day bamboo production training, the MMSU Extension Directorate and CAFSD worked with Barangay Aglipay and ILAARRDEC.
2. **THE RESULT**
   46 farmers, local leaders and community members established a new bambusetum, a collection of bamboo species, in the barangay.
3. **THE SKILLS**
   Bamboo propagation, nursery establishment and plantation management.

46	2	1
participants	training days	new bambusetum

LENS FOR 15.3.5

A barangay-hosted bamboo collection championed by a local Magsasaka Siyentista.

Source: MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025



Project Leader Dr. Arlene L. Gonzales presents the project at its launch, MMSU Review Center, 27 June 2024. (Image credit: TTPD, DOST-PCAARRD)



The MMSU project team with DOST-PCAARRD, DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao at the launch. (Image credit: TTPD, DOST-PCAARRD)

1 MAY 2024 – 30 APR 2026 · BADOC AND CURRIMAO

Mangrove Nature-based Solutions Project Runs from 2024 to 2026

Continuing implementation · 1 May 2024 to 30 April 2026

1 **THE PROJECT**
The DOST-PCAARRD-funded NbS project led by Dr. Arlene L. Gonzales launched on 27 Jun 2024 at the MMSU Review Center, supported by DENR-PENRO Ilocos Norte, the Provincial Government and the LGUs of Badoc and Currimao.

2 **THE PROBLEM**
Poor hydrological conditions keep young mangrove propagules from surviving.

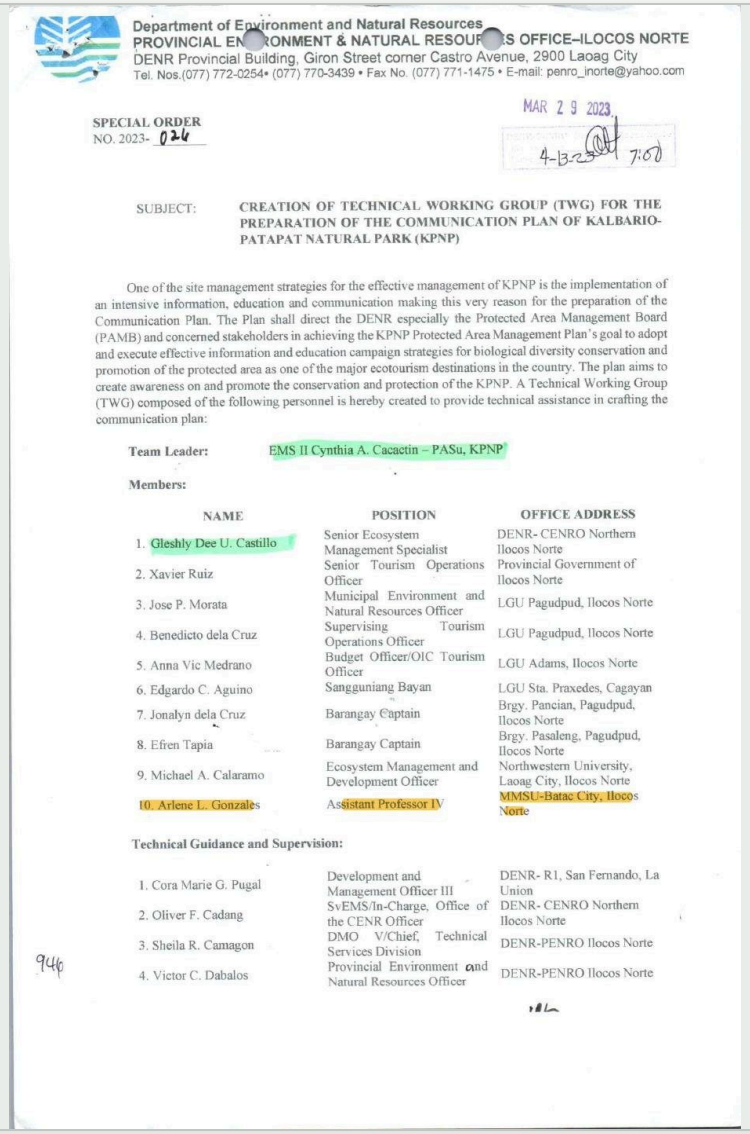
3 **THE FIX**
Technology-assisted plantations using locally available, ecologically friendly materials, including bamboo-engineered structures. The 2025 activities are shown in the 15.2.3 and 15.2.4 frames.

24	2
months of project term	municipalities

LENS FOR 15.3.5

Continuing: a multi-LGU, DENR-supported project throughout the evidence year.

Source: DOST-PCAARRD, “DOST-PCAARRD, MMSU launch nature-based solutions project on mangrove management in Ilocos Norte,” 7 August 2024



Department of Environment and Natural Resources
PROVINCIAL ENVIRONMENT & NATURAL RESOURCES OFFICE–ILOCOS NORTE
DENR Provincial Building, Giron Street corner Castro Avenue, 2900 Laoag City
Tel. Nos.(077) 772-0254• (077) 770-3439 • Fax No. (077) 771-1475 • E-mail: penro_inorte@yahoo.com

SPECIAL ORDER
NO. 2023- 024

MAR 29 2023

SUBJECT: CREATION OF TECHNICAL WORKING GROUP (TWG) FOR THE PREPARATION OF THE COMMUNICATION PLAN OF KALBARIO-PATAPAT NATURAL PARK (KPNP)

One of the site management strategies for the effective management of KPNP is the implementation of an intensive information, education and communication making this very reason for the preparation of the Communication Plan. The Plan shall direct the DENR especially the Protected Area Management Board (PAMB) and concerned stakeholders in achieving the KPNP Protected Area Management Plan's goal to adopt and execute effective information and education campaign strategies for biological diversity conservation and promotion of the protected area as one of the major ecotourism destinations in the country. The plan aims to create awareness on and promote the conservation and protection of the KPNP. A Technical Working Group (TWG) composed of the following personnel is hereby created to provide technical assistance in crafting the communication plan:

Team Leader: EMS II Cynthia A. Cacactin – PASu, KPNP

Members:

NAME	POSITION	OFFICE ADDRESS
1. Gleshly Dee U. Castillo	Senior Ecosystem Management Specialist	DENR- CENRO Northern Ilocos Norte
2. Xavier Ruiz	Senior Tourism Operations Officer	Provincial Government of Ilocos Norte
3. Jose P. Morata	Municipal Environment and Natural Resources Officer	LGU Pagudpud, Ilocos Norte
4. Benedicto dela Cruz	Supervising Tourism Operations Officer	LGU Pagudpud, Ilocos Norte
5. Anna Vic Medrano	Budget Officer/OIC Tourism Officer	LGU Adams, Ilocos Norte
6. Edgardo C. Aguino	Sangguniang Bayan	LGU Sta. Praxedes, Cagayan
7. Jonalyn dela Cruz	Barangay Captain	Brgy. Pancian, Pagudpud, Ilocos Norte
8. Efren Tapia	Barangay Captain	Brgy. Pasaleng, Pagudpud, Ilocos Norte
9. Michael A. Calaramo	Ecosystem Management and Development Officer	Northwestern University, Laoag City, Ilocos Norte
10. Arlene L. Gonzales	Assistant Professor IV	MMSU-Batac City, Ilocos Norte

Technical Guidance and Supervision:

1. Cora Marie G. Pugal	Development and Management Officer III	DENR- R1, San Fernando, La Union
2. Oliver F. Cadang	SvEMS/In-Charge, Office of the CENR Officer	DENR- CENRO Northern Ilocos Norte
3. Sheila R. Camagon	DMO V/Chief, Technical Services Division	DENR-PENRO Ilocos Norte
4. Victor C. Dabalos	Provincial Environment and Natural Resources Officer	DENR-PENRO Ilocos Norte

946

Document: DENR-PENRO Ilocos Norte Special Order No. 2023-024 (29 March 2023) naming Dr. Arlene L. Gonzales of MMSU to the KPNP Technical Working Group.

SPECIAL ORDER NO. 2023-024 · 29 MAR 2023 · DENR-PENRO ILOCOS NORTE

MMSU Expert Serves on the Kalbario-Patapat Natural Park Technical Working Group

Continuing role · appointment of 29 March 2023

1. **THE APPOINTMENT**
   Dr. Arlene L. Gonzales of MMSU was named to the Technical Working Group preparing the Communication Plan of the Kalbario-Patapat Natural Park, a protected area in Ilocos Norte.

2. **THE GROUP**
   Led by the KPNP Protected Area Superintendent, with DENR offices, the Provincial Government, the LGUs of Pagudpud, Adams and Sta. Praxedes (Cagayan), barangay officials and Northwestern University.

3. **THE GOAL**
   To support the Protected Area Management Plan's information and education campaigns for biodiversity conservation and the promotion of the park as a major ecotourism destination.

2023
appointment

3
partner LGUs

LENS FOR 15.3.5

Continuing role: MMSU sits on a protected-area working group with DENR, three LGUs and barangay officials.

Source: DENR-PENRO Ilocos Norte Special Order No. 2023-024, 29 March 2023 (MMSU SDG 15 page)

RECOMMENDED ANSWER ON THE THE FORM

Yes

2025 EVIDENCE

DRAFT COMMENT

670 / 1,000 characters

In 2025 MMSU collaborated with local communities to maintain shared land ecosystems: a memorandum of agreement with the TMP Foundation, DENR and LGU Pasuquin (29 Nov) for a degraded 2-ha coastal forest, where about 110 volunteers planted 1,750 mangrove seedlings with 3,500 more to follow; mangrove plantings with LGU Currimao, DENR and fisherfolk (250 propagules, Jun) and with the Rotary Club of Laoag Sunshine and LGU Badoc (280, 6 Dec); the MMSU–DENR Region 1 dialogue; indigenous-plant foodscapes with Vintar and San Nicolas partners; and a community bambusetum in Pinili. mmsu.edu.ph/news/mmsu-ties-up-with-toyota-and-denr-for-coastal-preservation-in-pasuquin-town

BEFORE YOU SUBMIT

- Attach a copy of the signed Pasuquin memorandum of agreement (29 Nov 2025).
- Confirm the date of the 2025 MMSU–DENR Region 1 dialogue at ILAARRDEC.
- Mangrove NbS project term ended 30 Apr 2026; the Kalbario-Patapat appointment dates from 2023. Both are continuing evidence.
- Confirm the 2027 data year with THE before upload.
- Upload one file per indicator (this storyboard is built for that). THE rejects evidence that is mainly a list of links.

EVIDENCE LINKS (10)

1. MMSU News, “MMSU ties up with Toyota and DENR for coastal preservation in Pasuquin town,” 4 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-toyota-and-denr-for-coastal-preservation-in-pasuquin-town
2. The Manila Times, “TMP Foundation adopts mangrove forest for climate mitigation,” 10 December 2025 https://www.manilatimes.net/2025/12/10/tmt-newswire/tmp-foundation-adopts-mangrove-forest-for-climate-mitigation/2239598
3. MMSU News, “MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town,” 11 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-rotary-club-of-laoag-sunshine-for-mangrove-planting-in-badoc-town
4. MMSU News, “MMSU plants 250 mangrove propagules in Currimao,” 11 June 2025 https://www.mmsu.edu.ph/news/mmsu-plants-250-mangrove-propagules-in-currimao
5. MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking no public link yet
6. MMSU News, “MMSU trains Ilocos Norte coastal towns on mangrove propagation and management,” 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management
7. MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
8. MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili
9. DOST-PCAARRD, “DOST-PCAARRD, MMSU launch nature-based solutions project on mangrove management in Ilocos Norte,” 7 August 2024 https://www.pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/dost-pcaarrd-mmsu-launch-nature-based-solutions-project-on-mangrove-management-in-ilocos-norte
10. DENR-PENRO Ilocos Norte Special Order No. 2023-024, 29 March 2023 (MMSU SDG 15 page) https://www.mmsu.edu.ph/resources/1TCpM-DxbkgfOh4QGwlfVr8QavCchKWKu/data